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## **Multimedia Appendix**

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## Multimedia Appendix 2

### Three-Pillar Organizational Assessment Rubric

This appendix presents the full Three-Pillar Assessment Rubric proposed in the main paper (section “The Three-Pillar Rubric”). Each indicator is scored as Low, Medium, or High Strength using evidence-based anchors from the literature reviewed in the main paper’s Evidence Base section. Organizations scoring predominantly “Low Strength” across multiple pillars face the self-reinforcing degradation cycle that the framework identifies as the central threat.

#### Pillar 1: Analytics Maturity

| Indicator              | Low Strength                                      | Med. Strength                                                 | High Strength                                       | Evidence |
|------------------------|---------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------|----------|
| HIMSS AMAM Stage       | Stages 0-2:<br>Fragmented data, limited reporting | Stages 3-4:<br>Integrated warehouse, standardized definitions | Stages 5-7:<br>Predictive analytics, AI integration | [1,2]    |
| Self-service analytics | None; all analytics require IT intervention       | Partial; BI tools available but underutilized                 | Widespread; clinical staff access data directly     | [3,4]    |
| AI/NL interface        | No NL2SQL or conversational analytics             | Pilot programs or evaluation underway                         | Natural language query capability deployed          | [5,6]    |

#### Pillar 2: Workforce Agility

| Indicator                           | Low Strength                                      | Med. Strength                              | High Strength                               | Evidence |
|-------------------------------------|---------------------------------------------------|--------------------------------------------|---------------------------------------------|----------|
| First-year analytics staff turnover | >30% (High instability)                           | 15-30%                                     | <15% (High stability)                       | [7,8]    |
| Avg. leadership tenure              | < 3 years                                         | 3-5 years                                  | > 5 years                                   | [9]      |
| Knowledge concentration             | Critical expertise held by 3 or fewer individuals | Partial documentation; some cross-training | Distributed expertise; documented processes | [10,11]  |

### Pillar 3: Technical Enablement

| Indicator        | Low Strength                                     | Med. Strength                                    | High Strength                                | Evidence |
|------------------|--------------------------------------------------|--------------------------------------------------|----------------------------------------------|----------|
| Data access      | SQL/technical expertise required for all queries | IT queue for complex queries; basic self-service | Natural language or visual query interfaces  | [4,6]    |
| Interoperability | Fragmented systems; manual reconciliation        | Partial integration; some automated feeds        | Unified data platform; real-time integration | [12,13]  |
| Schema coupling  | Hard-coded reports break on schema change        | Partial semantic layer; some automated feeds     | Semantic layer adapts; CI/CD detects drift   | [14,15]  |

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